

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

5th Semester of B. Pham Examination

University Theory Examination November 2017

PH319 Pharmaceutical Microbiology & Biotechnology

Date: 23/11/2017, Thursday Time:10:00 a.m. to 1:00 p.m. Maximum Marks:80

Instructions:

1. There are three sections in this question paper.
 2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
 3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
 4. Figures to right indicate marks.
 5. Draw neat sketches wherever necessary.
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SECTION - I

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|------------|--|-----------|
| Q 1 | Attempt all questions. Each question is of one mark. | 20 |
| 1. | What is the capacity of magnification of light microscope
A. 2,00,0 times
B. 1,00,0 times
C. 4,00,0 times
D. 40,000 times | |
| 2. | Which of the following is bacteriocidal method
A. Membrane filtration
B. Ionizing Radiation
C. Freeze drying
D. Deep freezing | |
| 3. | Which of the following item can be sterilized by dry heat sterilization.
A. IV Infusion liquids
B. Plastics IV bags
C. Glass pipettes
D. Rubber gloves | |
| 4. | Why microbial biotransformation is preferred over plant or animal cell culture?
A. Fast Growth rate
B. High metabolism rate
C. Large surface area
D. All | |
| 5. | The plasmid can be eliminated from a cell by the process known as _____.
A. Curing
B. Breking
C. Fixing
D. Expulsion | |

6. Which type of plasmid can exist with or without being integrated into the host's chromosome?
A. Medisome
B. Lisosome
C. Lysogen
D. Episome
7. Which of the following is the cause for drug resistance in TB?
A. Mutation
B. Transduction
C. Transformation
D. Conjugation
8. The correct term for the transfer of genetic material between bacteria in direct physical contact is _____
A. Mutation
B. Transduction
C. Transformation
D. Conjugation
9. F factor plasmids play a major role in _____
A. Mutation
B. Transduction
C. Transformation
D. Conjugation
10. From the following which microorganisms are able to grow at 0°C or lower temperature than 0°C?
A. Mesophiles
B. Thermophiles
C. Stenothermophiles
D. Psychrophiles
11. What is the full form of ATCC
A. American Type Culture Collection
B. American Type Collection of Culture
C. American Typical Culture Collection
D. American Types Culture Collection
12. Which of the following is not a type of light microscopy?
A. Bright-field
B. Dark-field
C. Phase-contrast
D. Fluorescence
13. Bactericide means
A. Killer of bacteria
B. Inhibitor of bacterial growth
C. Enhancer of bacterial growth
D. Inducer of bacterial growth
14. Autoclave is used for
A. Sterilization
B. Heating
C. Drying
D. Disinfection

15. An attenuated vaccine is composed of.
 - A. killed microorganisms
 - B. living, weakened microorganisms
 - C. inactivated bacterial toxins
 - D. recombinant vectors
16. Double-strand breaks in DNA are lethal to the descendants of
 - A. Bacteria
 - B. Insects
 - C. Fungi
 - D. Algae
17. Thermolabile pharmaceutical products can be sterilized by _____
 - A. Filtration
 - B. Dry heat sterilization
 - C. Moist heat sterilization.
 - D. Radiation sterilization.
18. Microbial assay is used for _____
 - A. To check turbidity
 - B. To check presence of microorganism
 - C. To check the efficacy of antibiotics
 - D. To check effect of phenol
19. Zone of _____ will be observed for one level microbial assay of vitamins
 - A. Exhibition
 - B. Inhibition.
 - C. Dispersion
 - D. Suspension
20. From the following which one is the biological indication for moist heat sterilization.
 - A. Bacillus stercorophilus
 - B. Bacillus subtilis
 - C. Bacillus pumilis
 - D. E. coli

SECTION – II

- Q 2** Attempt any TWO of the following
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|----------|---|-----------|
| A | Define pharmaceutical microbiology. Discuss the scope of microbiology in Pharmaceutical industry. | 05 |
| B | Classify the microorganisms. Differentiate the bacteria and viruses | 05 |
| C | Differentiate between bright field microscopy and electron microscopy techniques. | 05 |
- Q 3** Attempt any TWO of the following
- | | | |
|----------|---|-----------|
| A | Describe various methods used for isolation of microorganism. Compare its advantages & disadvantages. | 05 |
|----------|---|-----------|

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|----------|---|-----------|
| B | Enumerate the conditions requires for the growth of microorganism. Draw a typical bacterial growth curve. | 05 |
| C | Define gene cloning. How gene cloning can be usedul for the treatment of dieases. | 05 |

SECTION – III

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| Q 4 | Attempt any <u>FOUR</u> of the following | |
| A | Classify the disinfectants. Discuss the mechanism of action of any three. | 05 |
| B | Define disinfection. Classify the factors influencing disinfectant and discuss any two. | 05 |
| C | Define the following terms: 1. Disinfection 2. Thermal death time 3. Sterilization 4. Tyndallization 5. D value | 05 |
| D | Give the evaluation of Disinfectant. Explain Rideal-Walkar test. | 05 |
| E | Define sterilization. Classify the different methods of sterilization with mechanism of action. | 05 |
| F | Why test for sterility is carried out? What is the importance of control test in the test for sterility? | 05 |
| Q 5 | Attempt any <u>FOUR</u> of the following | |
| A | Classify various methods of sterilization. Describe working principle and applications of autoclave. | 05 |
| B | Explain immobilization techniques in detail. | 05 |
| C | Define replication and give a note on enzymes used in replication of DNA. | 05 |
| D | Write a note on Microbiological assay of Antibiotics. | 05 |
| E | Explain production of streptomicyne by fementation. | 05 |
| F | Enlity the types of microbial biotranformation and explain any one in detail. | 05 |